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FSTNet Frequency-Spatial Transformer Network for Camouflaged Object Detection

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Abstract

Camouflaged target detection (COD) is a challenging task in the field of computer vision, aiming to accurately segment and localize objects that are highly similar to their surroundings. Existing methods are often difficult to effectively capture the subtle differences between camouflaged targets and the background, and especially perform poorly in complex scenes. To address the above problems, a novel Frequency-Spatial Transformer Network (FSTNet) is proposed in this paper. The network first uses the pre-trained PVTv2 as the backbone network to extract multi-level features, and then captures rich information in both spatial and frequency domains simultaneously by the innovatively designed multi-scale feature aggregation (MSFA) module. In addition, a multi-scale transform fusion (MSTF) module is designed in this paper to efficiently fuse the feature representations at different scales by utilizing wavelet transform and Transformer decoder. Finally, the features are refined by Wavelet Enhanced Feature Integration Module (WFIM) to generate the final camouflaged target prediction results. Extensive experiments on four publicly available benchmark datasets show that the proposed FSTNet outperforms 10 existing state-of-the-art methods in several evaluation metrics, such as average absolute error, enhanced alignment metric, structure metric, and average F metric. In addition, detailed ablation experiments further validate the effectiveness of each key component of the network.

The visualization results show that the proposed method has significant advantages in dealing with challenging scenarios such as small objects, multiple objects, and objects highly similar to the background.

Keywords: COD, Image segmentation, Object detection

1 Introduction

It is evident that creatures in nature have evolved a remarkable capacity for camouflage, enabling them to blend seamlessly into their surroundings. This adaptation involves the ability to modify various physical attributes, such as shape, coloration, and texture, thereby reducing the probability of detection by predators and enhancing their chances of survival. The utilization of biological camouflage properties has found application in a variety of contexts, including the domains of clothing design and military reconnaissance. Furthermore, these properties have served as a catalyst for the advancement of technology in the field of computer vision, specifically the development of Camouflaged Object Detection (COD) technology. The primary objective of COD technology is to accurately segment camouflaged objects that exhibit a high degree of similarity to their background. This subject has garnered significant attention within the field of computer vision in recent years[1–5]. The efficacy of this technique has been demonstrated in a variety of practical applications, including medical image analysis [6–8], crop pest monitoring [9, 10], rare species discovery[11], search and rescue , and surface defect detection [12].

Early conventional methods mainly relied on low-level features for camouflaged target recognition, such as texture features[13], motion contours[14], optical flow features [15], and 3D convexity features[16]. However, camouflaged objects are able to dynamically adjust their colors and shapes to adapt to the surrounding environment, making it difficult for traditional methods to capture the subtle differences between the camouflaged object and the background, leading to poor recognition results.

With the development of deep learning technology, deep neural network-based COD methods show significant advantages. Fan et al.[17]inspired by the hunting behavior of predators in nature, proposed a SINet network consisting of a search module and a recognition module, and constructed a large-scale COD dataset, COD10K. Zhou et al.[18] proposed a Feature Aggregation and Propagation Network (FAP-Net) that learns boundary-enhanced representations through a layer-wise manner using Boundary Guidance Module (BGM) to help detect the boundaries of camouflaged objects. Fang et al.[19] proposed an Edge Perception in Frequency Domain Network (EPFDNet) that extracts edge information through deep wavelet-like transformation (DWT) using high-frequency components to accurately segment the edges of camouflaged objects. Recently, wavelet-based frequency decomposition has demonstrated remarkable success in various visual perception tasks. Zhang et al.[20] proposed a spatio-temporal wavelet attention aggregation network that effectively captures multi-scale temporal-frequency features for remote sensing change detection. Similarly, wavelet-guided dual-frequency encoding[21] has shown the advantages of

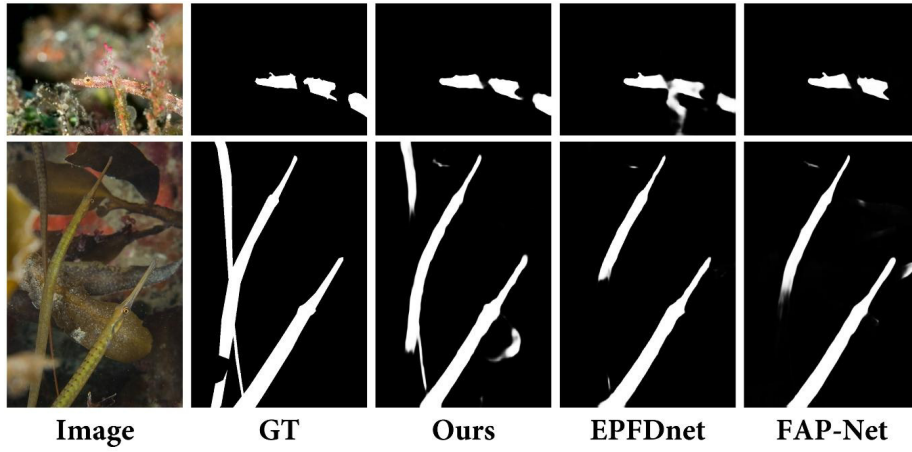


Fig. 1 Results of the above methods

jointly modeling high-frequency and low-frequency components for fine-grained visual discrimination.

Despite recent advances, existing methods struggle to achieve accurate and complete detection when camouflaged objects exhibit high similarity to their backgrounds. In complex environments with foreground occlusions, detection results frequently exhibit incomplete structures and blurred boundaries, failing to preserve fine structural details. As illustrated in Fig.1, state-of-the-art COD methods (e.g., EPFDNet and FAPNet) still produce incomplete segmentation when camouflaged objects are highly integrated with their backgrounds. These methods inadequately capture structural and boundary information in challenging scenarios, highlighting the need for more robust approaches to handle complex camouflage patterns.

Motivated by the aforementioned challenges, we propose FSTNet, a frequency-space fusion network for camouflaged object detection. Our network leverages PVTv2 as the backbone for multi-level feature extraction and incorporates three novel modules: Multi-Scale Feature Aggregation (MSFA), Multi-Scale Transform Fusion (MSTF), and Wavelet-enhanced Feature Integration Module (WFIM). The MSFA module fuses frequency and spatial domain features to address edge blurring issues. The MSTF module employs multi-scale transformation for comprehensive feature fusion. The WFIM module utilizes wavelet transform to enhance feature representation and integrate multi-stage features for final prediction. The main contributions of this paper are summarized as follows:

1. We propose FSTNet, a novel frequency-space fusion network that improves camouflaged object detection by separately processing and fusing frequency and spatial features through modular design.
2. We design two complementary modules to enhance detection performance:

MSFA: Employs Fourier frequency separation and multi-branch receptive fields to extract edge details and texture differences at multiple scales.

MSTF: Combines wavelet transform with Transformer architecture to fuse multi-scale features while preserving frequency-domain details.

WFIM: utilizes the wavelet transform to enhance the feature representation, integrates feature information from different processing stages, and simultaneously captures regional continuity features and edge texture differences through a three-path structure.

3. Extensive experiments on four benchmark datasets demonstrate that FSTNet outperforms state-of-the-art methods across five evaluation metrics, validating our approach’s effectiveness.

2 Related Work

2.1 Camouflage Object Detection COD

Camouflaged Object Detection (COD) seeks to localize and segment objects that are visually similar to their surroundings. Early approaches relied on handcrafted cues such as color, texture, shape, 3D convexity, and motion, but they struggle in cluttered scenes or when foreground and background are highly similar. With deep learning, performance has improved markedly. Fan et al.[17] proposed SINet, a two-stage search-and-identification framework, and released the large-scale COD10K dataset. Zhou et al.[18] iteratively decoupled body cues and integrated detail cues to form comprehensive representations. Beyond single-domain cues, recent work fuses spatial and frequency information. Zhu et al.[22] utilizes two-stream separated attention modules to highlight the camouflaged object’s boundary between an image’s background and foreground. Zhou et al.[23] iteratively decoupled body cues and integrated detail cues to form comprehensive representations. Beyond single-domain cues, recent work fuses spatial and frequency information. Hu et al.[24] modeled cross-band correlations with frequency self-attention. In a related domain, Chen et al.[25] proposed a motion and appearance decoupling representation that demonstrates the effectiveness of separating different visual cues for enhanced feature learning, which inspires our design of decoupled frequency-spatial processing branches. Nevertheless, detecting objects under extreme background similarity—low contrast, heavy clutter, and subtle boundaries—remains challenging due to texture-noise amplification and edge blurring.

2.2 Feature Fusion Mechanisms

An effective feature fusion mechanism can fully utilize the complementary nature of features from different sources and enhance the overall feature representation capability. In multimodal feature fusion, simple splicing or summing operations often fail to fully exploit the correlation between features. The introduction of the attention mechanism provides a new solution for feature fusion, which realizes adaptive feature integration by learning the weight assignment among features. Zhang et al.[26] demonstrated that adversarial spatio-temporal learning can effectively model the interactions

between spatial and temporal features, providing theoretical support for joint feature learning across different domains.

Recent advances in multi-modal and cross-domain feature fusion have shown promising results. Zhang et al.[27] proposed a dual-branch frequency-spatial joint perception cross-modality network (FSCMF) that effectively integrates frequency-domain and spatial-domain information through parallel processing branches, demonstrating the superiority of joint frequency-spatial modeling in visual fusion tasks. This paradigm of separately processing and then adaptively fusing frequency and spatial features has proven effective in capturing complementary information from different representation domains.

Inspired by the above studies, this paper proposes a novel frequency-space feature fusion framework. Our approach first decomposes the input features into high-frequency and low-frequency components using an improved wavelet transform frequency separation module to capture the edge texture information and the overall structure information, respectively. Then, a multi-branch structure is designed to process the original spatial features to capture multi-scale spatial information through different sizes of receptive fields. Finally, the frequency feature fusion module is used to adaptively fuse the output spatial features with the processed frequency features to form a more discriminative feature representation.

This design takes full advantage of the complementary nature of the frequency and spatial domain information: the high-frequency information helps to enhance the representation of boundary and texture details, the low-frequency information preserves the overall structural features of the target, and the multi-scale spatial features provide rich semantic information. Through the adaptive fusion of the FFM module, the network is able to dynamically adjust the contribution weights of different features according to the specific input content, thus realizing more accurate target detection in complex camouflage scenarios.

3 Methodology

3.1 FSTNet Network

Fig.2. illustrates the overall architecture proposed by the method in this paper. Specifically, we first use the pre-trained PVTv2 model as a backbone network to extract multilevel features from the input image to generate spatial and semantic cues, $f_i (i = 1, 2, 3, 4)$. In the perception stage, the network integrates the multilevel features with the MSFA module for feature integration to extract richer semantic information about the features $F_i (i = 1, 2, 3, 4)$, thus extracting the features. In Addition, the MSTF module is utilized to perform feature fusion of the enriched features to obtain $G_i (i = 1, 2, 3, 4)$ as a priority guidance. Finally, in the feature refinement stage, we use WFIM to enhance the feature representation using the wavelet transform, integrate feature information from different processing stages, and obtain iterative features by capturing both the regional continuity features and the edge texture differences through a three-path structure and the topmost WFIM output G_i is used as the final pre-diction map. In the following summary, we accept the key modules and loss functions in FSTNet in detail.

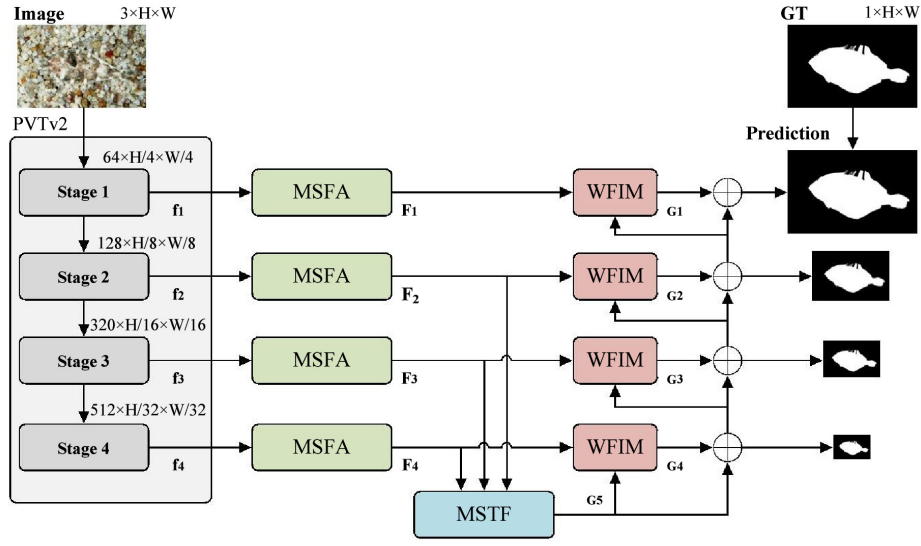


Fig. 2 FSTNet Network

3.2 MSFA Module

Given that camouflaged targets are sensitive to morphological changes in appearance, extracting contextual information through the near-neighborhood environment of camouflaged objects is important to improve effective target recognition. The structure of our proposed MSFA is shown in Fig. 3. It consists of three basic components: the MSFAU (Multi-Scale Feature Aggregation Unit), the FFU (Frequency Fusion Unit), and the FFM (Feature Fusion Module). The process is defined as follows:

$$\begin{aligned}
 F_{MSFAU}^1 &= Dconv_{3 \times 3}(Conv_{3 \times 1}(Conv_{1 \times 3}((Conv_{1 \times 1}(f_x)))))) \\
 F_{MSFAU}^2 &= Dconv_{5 \times 5}(Conv_{5 \times 1}(Conv_{1 \times 5}((Conv_{1 \times 1}(Cat(f_x, F_{MSFAU}^1)))))) \\
 f_b &= Cat(F_{MSFAU}^1, F_{MSFAU}^2) \\
 f_a &= Conv_{1 \times 1}(f_x)
 \end{aligned} \tag{1}$$

Where F_{MSFAU}^1 and F_{MSFAU}^2 are the two branches of the MFAU, $Conv_{k \times k}$ represents the normal convolution with $k \times k$ convolution kernels and $Dconv_{k \times k}$ represents the dilated convolution with $k \times k$ convolution kernels. Finally, the results of the two branches are merged by utilizing the channel linking operation and the output is f_b , f_a is the other output of the module, and $Conv_{1 \times 1}$ is used for the output of the next module after feature f_x extraction through the pair. The frequency domain fusion unit FFU aims to extract the spatial-frequency features of the object

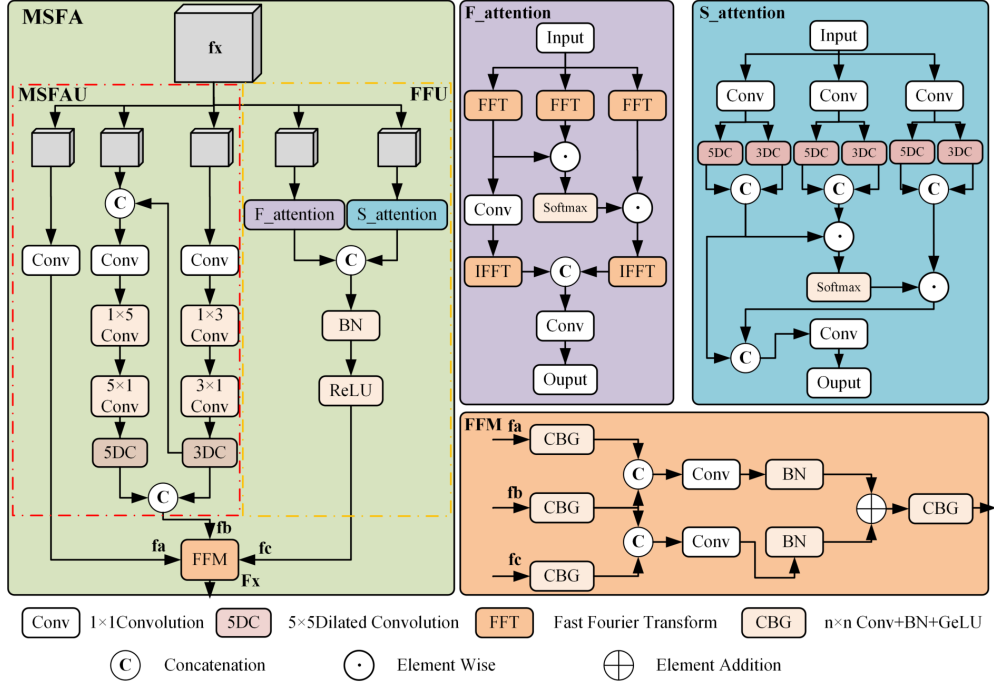


Fig. 3 Overview of the Structure of MSFA

through the attention mechanism in both spatial and frequency domains to improve the features. The features are first extracted by two different attention modules and then the extracted features are spliced using the channel splicing operation. And the output is obtained by batch normalization and ReLU activation function processing f_c . The process is defined as follows:

$$f_c = F_{FFU} = \text{ReLU}(\text{BN}(\text{Cat}(F_{\text{attention}}(f_x), S_{\text{attention}}(f_x)))) \quad (2)$$

Where $F_{\text{attention}}$ is the frequency attention module gets the output by Fourier transform and Fourier inverse transform pairs, f_c standard activation units (BN and ReLU). The process is defined as follows:

$$\begin{aligned} F_{\text{attention}}^1 &= \text{IFFT}(\text{Conv}_{1 \times 1}(\text{FFT}(f_x))) \\ F_{\text{attention}}^2 &= \text{IFFT}(\text{Softmax}(\text{FFT}(f_x) \odot \text{FFT}(f_x)) \odot \text{FFT}(f_x)) \\ F_{\text{attention}}^{\text{out}} &= (\text{Conv}_{1 \times 1}(\text{Cat}(F_{\text{attention}}^1, F_{\text{attention}}^2))) \end{aligned} \quad (3)$$

Where $F_{\text{attention}}^1$ and $F_{\text{attention}}^2$ are the two branches of $F_{\text{attention}}$. representing the Fourier feature extraction, representing the inverse Fourier feature extraction and the activation function. The frequency characteristics of the target are extracted by simple transform and inverse transform, and the frequency characteristics of the target

are extracted more deeply by interconnecting the multi-level transform and inverse transform with the activation function, and finally the results of the two branches are merged by utilizing the channel splicing operation and are outputted through the channel splicing operation. $S_{attention}$ is the spatial attention mechanism, which extracts the target features by multilayer dilated convolution and obtains the extracted target features by mutual fusion between the features. The process is defined as follows:

$$\begin{aligned} S'_{attention} &= Conv_{1 \times 1}(Cat(Dconv_{5 \times 5}(Conv_{1 \times 1}(f_x)), Dconv_{3 \times 3}(Conv_{1 \times 1}(f_x)))) \\ S^1_{attention} &= S'_{attention} \\ S^2_{attention} &= (Softmax(S'_{attention} \odot S'_{attention}) \odot S'_{attention}) \\ S^{out}_{attention} &= (Conv_{1 \times 1}(Cat(S^1_{attention}, S^2_{attention}))) \end{aligned} \quad (4)$$

Where $S^1_{attention}$ and $S^2_{attention}$ are the two branches of $S_{attention}$, and $S'_{attention}$ is the intermediate layer, the target is subjected to multi-scale feature extraction by different convolutional kernel sizes of dilated convolution and different convolutions, and the features are fused by similar $F^2_{attention}$ in-ter-feature fusion operation. Finally, the results of the two branches are merged by utilizing the channel splicing operation and output through the $Conv_{1 \times 1}$.

The feature fusion module performs feature fusion of the parts of each module, and the features are fused and extracted by splicing and summing operations between the features. The process is de-fined as follows:

$$\begin{aligned} F^1_{FFM} &= BN(Conv_{1 \times 1}(Cat(CBG(f_a), CBG(f_b)))) \\ F^2_{FFM} &= BN(Conv_{1 \times 1}(Cat(CBG(f_b), CBG(f_c)))) \\ F^{out}_{FFM} &= CBG(F^1_{FFM} \oplus F^2_{FFM}) \end{aligned} \quad (5)$$

Where CBG is the joint operation of Conv, BN and GeLU modules, the purpose is regard-ed as the inter-feature channel alignment and pre-fusion operations on the features, and finally the final output features are obtained by summing the inter-feature elements, which F^{out}_{FFM} is also the output of the MSFA module.

Although the MSFA module incorporates multiple parallel branches and FFT-based operations, its computational overhead remains manageable for the following reasons: (1) The FFT/IFFT operations have a complexity of $O(HW \log(HW))$ at feature resolution $H \times W$, which is significantly lower than the $O(H^2W^2)$ complexity of global self-attention. (2) We apply 1×1 convolutions before and after each branch to reduce the channel dimensions, limiting the intermediate computation. (3) The MSFAU adopts asymmetric convolutions ($1 \times k$ and $k \times 1$) followed by dilated convolutions, which effectively enlarge the receptive field while keeping the parameter count low compared to standard large-kernel convolutions.

3.3 MSTF Module

To fuse semantic information across feature levels, we propose the MSTF module, which combines wavelet-based downsampling with a Transformer decoder, as shown in Fig. 4.

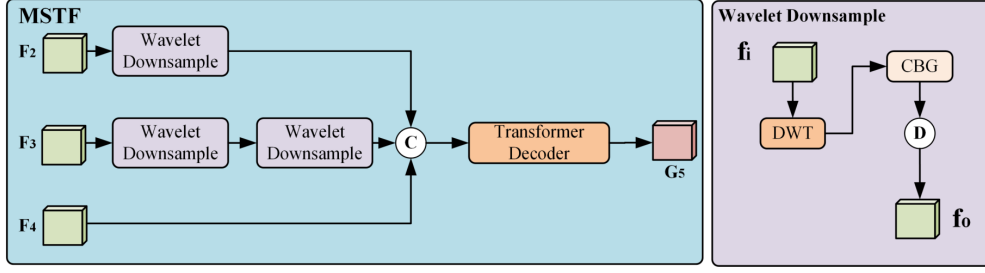


Fig. 4 Overview of the Structure of MSTF

We extracted the frequency detail features $F_i (i = 2, 3, 4)$ by performing wavelet decomposition downsampling transforms on the outputs of the three MSFA modules, which were then spliced-together and then fused using the Transformer Decoder module in order to improve the interaction between the layers. We adopt wavelet-based downsampling because DWT provides a joint spatial–frequency representation with strong locality, which helps preserve subtle boundary/edge cues that are crucial for camouflaged target detection while aligning multi-level features to a consistent spatial scale. The process of the MSTF module is defined as follows:

$$F_{MSTF} = \text{Transformer Decoder}(\text{Cat}(\text{WD}(F_2), \text{WD}(\text{WD}(F_3)), F_4)) \quad (6)$$

where WD denotes the wavelet decomposition downsampling operation, defined as

$$f_o = \text{Downsample}(\text{CBG}(\text{DWT}(f_i))). \quad (7)$$

Specifically, $\text{DWT}(\cdot)$ decomposes the input feature into different frequency components (sub-bands). These components are fused and refined by $\text{CBG}(\cdot)$ (Conv+BN+GeLU), and then $\text{Downsample}(\cdot)$ reduces the spatial resolution to facilitate multi-level feature alignment for subsequent fusion.

The concatenated features are fused by the Transformer decoder, which consists of a Multi-Head Self-Attention (MSA) sublayer and an MLP sublayer. Residual connections are applied to both sublayers to preserve feature integrity while enhancing cross-level information exchange. The fused representation is further aligned and pre-fused by $\text{CBG}(\cdot)$, and the final output is obtained via element-wise summation, which serves as the output of the MSTF module.

3.4 WFIM Module

Since there is a high degree of similarity between the camouflaged object and its environment in complex environments, the WFIM module proposed in this paper refines the camouflaged area and obtains the correlation between various features through iteration to obtain the location of the cam-ouflaged target. We introduce wavelet decomposition in WFIM to explicitly inject localized frequency cues into the refinement process: DWT highlights scale-dependent high-frequency responses while

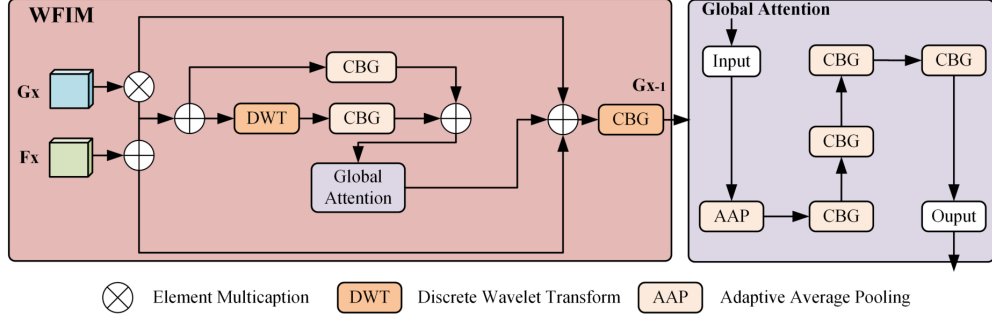


Fig. 5 Overview of the Structure of WFIM

preserving spatial localization, which helps suppress background clutter and emphasize camouflaged target boundaries. The structure of WFIM is shown in Fig. 5, in which we first interact G_x and F_x to obtain diverse forms of information through multiplication and addition. After that, we further refine the features through the global attention mechanism by wavelet decomposition and summing, and obtain the final output through the CBG module by summing the features G_{x-1} , and the process of WFIM module is defined as follows:

$$\begin{aligned}
 F_{WFIM}^1 &= G_x \otimes F_x \\
 F_{WFIM}^2 &= G_x \oplus F_x \\
 F_{WFIM}^3 &= GA(CBG(F_{WFIM}^1 \oplus F_{WFIM}^2) \oplus CBG(DWT(F_{WFIM}^1 \oplus F_{WFIM}^2))) \\
 F_{WFIM} &= CBG(F_{WFIM}^1 \oplus F_{WFIM}^2 \oplus F_{WFIM}^3)
 \end{aligned} \tag{8}$$

Among them, Global Attention is the global attention module, and its specific structure is defined as follows:

$$F_{GA} = CBG_4(CBG_3(CBG_2(CBG_1(AAP(f_i))))) \tag{9}$$

where f_i is processed by four cascaded CBG blocks, and $AAP(\cdot)$ denotes Adaptive Average Pooling to enhance robustness to varying feature sizes.

3.5 Loss Function

Following the traditional COD approach, our method employs binary cross entropy (BCE) loss and intersection and Intersection over Union (IoU) loss as the basic optimization tools in the field of image segmentation. However, the loss function affects each pixel uniformly and lacks the ability to dynamically assign individual pixel weights. We use wBCE and wIoU to calculate the difference between the center pixel and its surrounding region, so the total loss can be expressed as:

$$L = L_{BCE}^W + L_{IoU}^W \tag{10}$$

where L_{BCE}^W denotes weighted BCE loss and L_{IoU}^W denotes weighted IoU loss.

4 Experiment

In this section, we provide a detailed description of the experimental setup, followed by an in-depth elaboration of the dataset and evaluation metrics. We then compare our proposed FSTNet with other state-of-the-art methods. Finally, we conduct ablation experiments to validate the effectiveness of each key module.

4.1 Experiment Setting

The model in this paper is based on the PyTorch framework [28] and accelerated using NVIDIA RTX4080S GPUs. In the training phase, we use the PVTv2 backbone network pre-trained on ImageNet for initial parameter configuration. Also, the input images were resized to 416×416 pixels and a data enhancement strategy was applied to reduce the risk of overfitting in the model training. Optimization of FSTNet was performed by using the Adam optimization algorithm [29]. The model was trained for 100 epochs with the initial learning rate set to $1e-4$. In addition, the learning rate was reduced by a factor of 10 for every 40 epochs, and the batch size was set to 14. We evaluated the per-formance of the model after each iteration by using the COD test dataset to determine the optimal network weights.

4.2 Dataset

In order to evaluate the effectiveness of our FSTNet model, we conducted comparative experiments using four widely used benchmark data. The detailed information is as follows.

The CAMO dataset [30] contains 1250 camouflaged images categorized into eight different types. In this paper, 1000 of these images are selected for training and the remaining 250 images are used for testing.

The CHAMELEON dataset [31] is a small dataset labeled with the keyword “camouflage animal” in Google search engine, containing 76 images, all of which are used for testing.

The COD10K dataset [17] is a large camouflaged target dataset, including 5 categories and 69 subcategories, with a total of 5,066 images containing camouflaged objects. Of these, 3040 were used for training and 2026 images were used for testing.

The NC4K dataset [32] is the largest COD test dataset available and includes 4121 images containing camouflaged objects downloaded from the Internet.

The training set we used contains a total of 4040 images from CAMO (1000 images) and COD10K (3040 images). The test set we used contains a total of 6473 images from CAMO (250 images), CHAMELEON (76 images), COD10K (2026 images), and NC4K (4121 images).

4.3 Evaluation Indicators

We employ five widely used evaluation metrics in COD. These include Mean Absolute Error MAE, Enhanced Alignment Metric E_φ , Structural Metric S_α , and Mean F Metric F_β . MAE calculates the pixel-level average error between the predicted value

and the true value, and can be described as:

$$MAE = \frac{1}{W \times H} \sum_{i=1}^W \sum_{j=1}^H |P(i, j) - G(i, j)| \quad (11)$$

where W and H denote the width and height of G , respectively, and the value of MAE is normalized to $[0,1]$. Enhanced Alignment Metric E_φ is a foreground-based perceptual metric that takes into account the local and global similarity between predicted and true values. It can be described as:

$$E_\varphi = \frac{1}{W \times H} \sum_{i=1}^W \sum_{j=1}^H \varphi_{FM}(x, y) \quad (12)$$

where φ_{FM} is the enhanced alignment matrix. It evaluates the accuracy of the prediction globally and locally. In this paper we use E_φ^{mean} values as metrics. Structural metric S_α Emphasizes the structural similarity between predicted and true values, which can be described as:

$$S_\alpha = \alpha S_o + (1 - \alpha) S_r \quad (13)$$

where α is the balance factor between the control object perceptual recognition S_α and the region perceptual similarity S_r , which α is set to 0.5 based on previous work. the average F metric F_β is the weighted average of the measures of prediction precision (P) and recall (R). It is defined as:

$$F_\beta = \frac{(\beta^2 + 1)PR}{\beta^2 P + R} \quad (14)$$

4.4 Comparison with Other Models

In order to demonstrate the effectiveness of the proposed method, we compared the method with 10 state-of-the-art COD models using four evaluation metrics on four benchmark datasets including SINet [17], BSANet [22], C2FNet [33], OCENet [34], EPFDNet [19], FAPNet [35], BGNet [36], SINetV2 [37], SegMaR [29], and AFINet [38]. For a fair comparison, the experimental results of all the above methods were obtained either using a unified dataset and hyperparametric reconstruction based on the published source code, or using the authors' published prediction maps.

4.5 Quantitative evaluation

Table.1. summarizes the quantitative assessment results of our method on 11 COD tasks in SOTA on the four benchmark datasets. It can be seen that our method consistently leads on the four as-sessment metrics compared to other competitors. Analysis of CAMO Table 1 depicts the evaluation results of different models on the CAMO dataset. It is clear that FSTNet effectively optimizes the completeness of camouflage target generation by integrating multiple forms of feature information. When quantitatively comparing AFINet, SegMaR improves the scores on S_α , E_φ and F_β by 5.1%,

Table 1 Performance comparison of different methods on camouflage detection datasets.

Method	CAMO				CHAMELEON				COD10K				NC4K			
	S_α	MAE	E_φ	F_β	S_α	MAE	E_φ	F_β	S_α	MAE	E_φ	F_β	S_α	MAE	E_φ	F_β
SINet	0.751	0.100	0.831	0.706	0.869	0.044	0.935	0.830	0.771	0.051	0.868	0.676	0.808	0.058	0.883	0.775
BSANet	0.794	0.079	0.867	0.770	0.895	0.027	0.957	0.877	0.818	0.034	0.901	0.753	0.841	0.048	0.907	0.817
C2FNetv2	0.799	0.077	0.869	0.779	0.893	0.028	0.957	0.876	0.811	0.036	0.896	0.742	0.840	0.048	0.904	0.814
OCENet	0.802	0.080	0.865	0.777	0.897	0.028	0.955	0.870	0.827	0.033	0.905	0.764	0.853	0.045	0.913	0.831
EPFDNet	0.802	0.071	0.873	0.789	0.887	0.030	0.954	0.868	0.822	0.032	0.905	0.768	0.847	0.044	0.915	0.833
FAPNet	0.811	0.088	0.838	0.734	0.872	0.036	0.920	0.813	0.811	0.039	0.874	0.710	0.842	0.051	0.890	0.756
BGNet	0.812	0.073	0.882	0.799	0.901	0.027	0.954	0.881	0.831	0.033	0.911	0.774	0.851	0.044	0.916	0.833
SINetV2	0.814	0.072	0.873	0.777	0.891	0.029	0.941	0.840	0.814	0.037	0.884	0.718	0.847	0.047	0.902	0.805
SegMaR	0.815	0.071	0.884	0.803	0.892	0.027	0.962	0.861	0.833	0.034	0.906	0.774	0.841	0.046	0.907	0.826
AFINet	0.826	0.072	0.876	0.791	0.894	0.028	0.930	0.839	0.834	0.034	0.886	0.746	0.854	0.047	0.901	0.810
Ours	0.868	0.047	0.933	0.861	0.907	0.023	0.975	0.893	0.859	0.026	0.937	0.814	0.878	0.036	0.935	0.861

Table 2 Runtime and complexity analysis on RTX 4080S:Params(M),FLOPs(G) and FPS

Method	Params(M)	FLOPs(G)	FPS
SINet	48.90	19.50	103.0
BSANet	87.06	67.0	97.3
C2FNet	26.17	13.20	88.5
OCENet	54.98	25.30	98.6
EPFDNet	27.84	24.10	48.0
BGNet	77.80	58.50	91.4
SINetV2	24.93	12.31	106.8
SegMaR	55.62	33.65	130.9
AFINet	25.83	24.79	80.4
Ours	34.66	18.75	98.8

6.5%, and 8.9%. For CHAMELEON in Table.1, grids illustrate that our method consistently achieves the highest scores on the CHAMELEON dataset when compared to the 11 current methods. Compared to FAPNet, the scores on S_α , E_φ and F_β are improved by 4.0%, 6.0%, and 9.8%. Analysis of COD10K shows that our method exhibits its significant advantages on the very demanding COD10K dataset. Specifically, the scores on S_α , E_φ and F_β are improved by 3.0%, 3.3%, and 4.9%. For the analysis of NC4K we examined the performance scores of all methods on the entire test set, and our FSTNet showed a significant advantage. Table.1. shows that the value S_α is 3.2% higher than that of BGNet, and similarly, the value is 4.4% higher than that of SegMaR. The above analysis validates the effectiveness of our perceptual and iterative strategy for COD.

Besides accuracy, we evaluate computational complexity and inference speed in Table 2 and Fig. 6. Our model runs at 98.8 FPS on an RTX 4080S with 416×416 inputs, using 34.66M parameters and 18.75G FLOPs. As shown in Fig. 6, our method achieves a favorable trade-off between speed and complexity: compared with heavier models such as BSANet (87.06M, 67.0G) and BGNet (77.80M, 58.50G), it substantially reduces both parameters and FLOPs while maintaining competitive throughput.

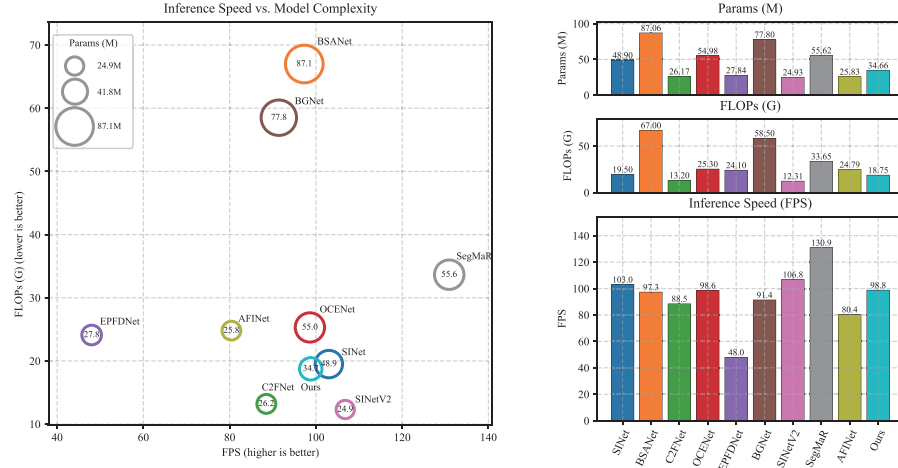


Fig. 6 Efficiency comparison of different COD models on RTX 4080S with 416×416 inputs. Left: FLOPs (G) versus inference speed (FPS), where the ring size indicates the number of parameters (M) and the value inside each ring denotes the corresponding Params (M). Right: bar charts of Params (M), FLOPs (G), and FPS.

Meanwhile, relative to lightweight baselines with lower complexity, our method provides a more balanced profile by retaining high inference speed together with strong overall performance (Table 1), which makes it more suitable for practical deployment.

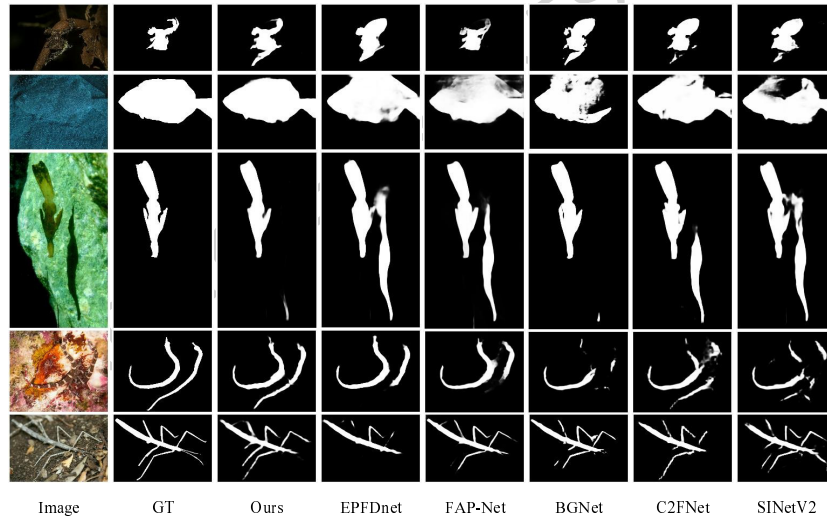


Fig. 7 Visual comparison of FSTNet, compared to other methods FSTNet is able to explore various camou-flaged objects in challenging scenarios

4.6 Qualitative evaluation

To further demonstrate the advantages of FSTNet, we have compiled visual comparison results from other SOTA methods, as shown in Fig.7.

4.7 Ablation experiments

Table 3 Performance comparison of different methods on camouflage detection datasets.

Method	CAMO				CHAMELEON				COD10K				NC4K			
	S_α	MAE	E_φ	F_β	S_α	MAE	E_φ	F_β	S_α	MAE	E_φ	F_β	S_α	MAE	E_φ	F_β
Basic	0.814	0.073	0.840	0.781	0.894	0.028	0.958	0.876	0.845	0.029	0.923	0.800	0.868	0.039	0.926	0.851
+MSFA	0.816	0.067	0.862	0.822	0.904	0.025	0.962	0.885	0.851	0.028	0.934	0.801	0.876	0.039	0.935	0.859
+MSTF	0.821	0.054	0.852	0.846	0.905	0.024	0.968	0.887	0.855	0.027	0.935	0.810	0.874	0.037	0.934	0.859
+WFIM	0.824	0.061	0.853	0.852	0.904	0.025	0.962	0.885	0.851	0.028	0.934	0.801	0.876	0.039	0.935	0.859
Ours	0.868	0.047	0.933	0.861	0.907	0.023	0.975	0.893	0.859	0.026	0.937	0.814	0.878	0.036	0.935	0.861

To validate the effectiveness of each core component of our FSTNet, the ablation results for the four baseline datasets are shown in Table.3. The effectiveness of the subcomponents was explored through decoupling operations.

we can observe that the performance of “Basic + MSFA” is lower than the feature representation of the overall network, which suggests that MSTF and WFIM can effectively integrate features of different resolutions through iterations. In addition, we confirm the importance of MSTF in the table, where experiments show that MSTF helps FSTNet to better focus on the target region and improves the visual recognition of camouflaged objects by as-signing different attentional weights. Correlation with better inference results. To explore the contribution of MSFA to COD performance,

Table 4 MSFA Module Ablation Experiment

Method	CAMO				CHAMELEON				COD10K				NC4K			
	S_α	MAE	E_φ	F_β	S_α	MAE	E_φ	F_β	S_α	MAE	E_φ	F_β	S_α	MAE	E_φ	F_β
-MSFAU	0.865	0.049	0.931	0.856	0.902	0.024	0.971	0.882	0.857	0.026	0.936	0.810	0.877	0.036	0.933	0.864
-FFU	0.863	0.052	0.928	0.856	0.904	0.025	0.962	0.885	0.851	0.028	0.934	0.801	0.876	0.039	0.935	0.859
-FFM	0.865	0.049	0.931	0.860	0.905	0.024	0.968	0.887	0.855	0.027	0.935	0.810	0.874	0.037	0.934	0.859
Ours	0.868	0.047	0.933	0.861	0.907	0.023	0.975	0.893	0.859	0.026	0.937	0.814	0.878	0.036	0.935	0.861

in Table.4. we performed ablation experiments using each of the four variants. The performance degradation in the “without FFU module” condition suggests that the null-frequency attention mechanism plays an important role in feature extraction. The performance degradation in “without MSFAU module” confirms that the use of convolutional kernels with different scales to capture features with different orientations and scales can improve the representation of local features, and thus enhance the overall performance of FSTNet. The degradation of “without FFM module” confirms

the effectiveness of merging features through multiple branches and fusion merging of features to improve the feature intersection.

5 Conclusion

In this paper we propose a new FSTNet network to achieve effective detection of camouflaged targets in different scenarios. Specifically, we firstly adopt the Transformer-based network to obtain the multilevel features of COD, in addition, the MSFA module is designed to efficiently fuse the frequency domain features with the spatial domain features via to deal with the problems such as blurred edge details in the camouflage target detection task. Then we designed the MSTF module, which utilizes the down-sampling module of waveform decomposition combined with the Transform Decoder to fuse the multilevel features generated by the previous module to improve the target localization and detection accuracy. Finally, we designed the WFIM module to refine the correlation between various features to obtain iteratively optimized camouflage masks. Extensive implementations on four benchmark datasets have shown that our proposed FSTNet outperforms 10 SOTA methods on four widely used evaluation metrics.

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Declarations

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Conflict of interest

The authors declare that they have no competing interests.

Ethics approval and consent to participate

Not applicable

Consent for publication

Not applicable.

Data availability

The datasets used and analysed in this study are fully publicly available online.

Author contribution

All authors take part in the discussion of the work described in this paper. These authors contributed equally to this work.

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